

# Analysis of the association between metaverse-based engineering education and environment, social and governance platforms

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## ABSTRACT

This paper proposed a metaverse environment-based engineering education method to utilize it as a platform that meets sustainable environmental, social, and governance according to social movements for reducing energy consumption and carbon footprint can be reduced by replacing them. Verifying the proposed method, we measured potential carbon footprint reduction as environmental factors and immersion as social and governance factors. Result of the experiment, the average value of all items was about 4.51, and the five items with the highest average value were separated from the real world due to the high sense of immersion and presence from the environment.

**Keywords:** Immersive Technology, Metaverse, Virtual Reality, Environmental, social and governance, Ethical issues.

**Index Terms:** J.4 [Computer Applications]: Social and behavioral sciences; K.4.2 [Computing Milieux]: Computers and society—Social issues

## 1 INTRODUCTION

The development of ICT technology has brought about many changes in various fields of human experience, especially in education, new teaching methods utilizing various new technologies that can improve the efficiency and effectiveness of learning are emerging [1]. Especially after the COVID-19 pandemic, a large part of human activities being transferred to digital platforms has become more frequent, and many people have experienced a revolution in their relationship with society and culture [2]. These phenomena have brought about changes in aspects of economic activities, forms of communication, and new and diverse areas of participation, the most representative of which is education. Recently, the field of education has shown a direction in line with social movements such as Education for Sustainable Development (ESD) and Environment, Social, Governance (ESG). Such trends are not only due to issues such as environmental sustainability, such as global warming and climate change but also to solving issues such as educational gaps, limited accessibility of educational services, unsafe environments, social discrimination and marginalization. Current education is centered around data, and within this education, data literacy has evolved beyond the ability to process data on one's own to include the role of ideologies such as equality, freedom, and ethics. Moreover, environmental concerns, such as claims that environmental issues, including global warming, caused viruses like COVID-19, have accelerated

the use of online platforms. This phenomenon has led to exponentially more data being generated in online environments, creating an educational gap due to data accessibility and efficiency of processing methods. Educational isolation caused by reduced social connectivity also deepened, strengthening negative outcomes such as learning effectiveness and motivation. The most crucial feature of immersive technology is that it motivates students to experience the virtual world with a high degree of freedom without space-time constraints, allowing them to experience new technology naturally. This experience can improve familiarity with the technology, increase interest, and encourage students to participate actively. After the global financial crisis and a series of natural disasters caused by climate change became global issues, ESG became a significant trend. Environmental standards can include energy consumption, environmental pollution, carbon emissions, and other ways to protect the environment, including various policies to address climate change. Social criteria consider managing relationships with society within the local environment in which it operates, such as communities, social relationships, and user interactions. Governance standards consider reducing social gaps within a given local environment and managing sustainability. Concerns about energy consumption and carbon emissions arise from concerns about the power generated during computing transactions. Regarding environmental standards, metaverse technology can reduce energy consumption and carbon emissions by replacing real-world and physical educational environments with interactive, digitized educational environments. The digital twin world created through migration as a metaverse environment helps optimize the real-world environment and helps overcome user behavioral barriers through free experience within the virtual world. In short, resource-intensive physical educational environments and real-world experiences in metaverse can be replaced with digital and virtual educational environments combined with visualized real-world, real-time data. These characteristics mean that replacing nature-intensive real-world goods and experiences in a metaverse environment with digital alternatives in a virtual environment can reap considerable benefits in terms of sustainability. Moreover, these virtual experiences are far less resource-intensive and carbon-efficient than real-world equivalents [3].

## 2 EVALUATION

We examined two factors to verify the usefulness of metaverse-based education as an ESG platform. The first is the carbon footprint, which can be reduced through verification through seminars in the metaverse environment. Figure 1 shows an online seminar conducted in a metaverse-based environment. The 340 students who attended the seminar connected through full-body avatars based on their appearance in a given virtual environment and engaged in non-face-to-face learning, such as senior mentoring education, infectious disease prevention, and safety education. Tables 1 and 2 show the results of calculating the carbon footprint generated by transportation and disposable items generated if a seminar was held offline in the metaverse environment as shown in

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Figure 1: Captured image for VR-based seminar using commercial application (Application: Spatial).

Figure 1. Through these results, the carbon emissions that the metaverse-based environment generated in this way could potentially reduce is 8,481 kgCO<sub>2</sub>eq, requiring approximately 1,437 cypress trees and 932 pine trees to purify. Therefore, metaverse environment-based education scheme can be proposed as a sustainable, environment-friendly training scheme in one way with an ecosystem of energy usage, pollution, and carbon footprint. The second is social interaction through the survey results of the seminar held in the VR-based metaverse environment. The 40 students who attended the virtual reality-based seminar also connected through full-body avatars created based on their images. The students then learned about the theory through presentations and used interactive three-dimensional models to carry out practical training in which what they had learned was realized through simulations. Table 3 summarizes the responses to the immersion evaluation questionnaires [4]. The factors that most significantly impact educational effectiveness in metaverse-based environments are explained by immersion and presence. This questionnaire measures the task, dissociation from the real world due to loss of awareness of the surrounding environment, emotional engagement, including attachment to virtual reality content, and cognitive engagement, indicating the degree of effort and concentration.

Table 1: Carbon footprint according to transportation.

| Transportation | Distance<br>km | Emission Factor<br>Kg/unit*km | Carbon Emission<br>kgCO <sub>2</sub> eq |
|----------------|----------------|-------------------------------|-----------------------------------------|
| Train          | 45,339         | 0.041                         | 1,861.35                                |
| Bus            | 58.567         | 0.105                         | 6,149.57                                |

Table 2: Carbon footprint according to disposable items for offline participants.

| Item           | MATL No. | WGT<br>Kg | Prod.<br>kgCO <sub>2</sub><br>eq/kg | Incineration<br>kgCO <sub>2</sub> eq<br>/kg | Carbon<br>Emission<br>kgCO <sub>2</sub> eq |
|----------------|----------|-----------|-------------------------------------|---------------------------------------------|--------------------------------------------|
| Cup            | PET 340  | 0.012     | 2.17                                | 1.24                                        | 13.91                                      |
| Water bottle   | PET 340  | 0.014     | 2.17                                | 1.24                                        | 16.23                                      |
| Snack          | PP 680   | 0.002     | 1.52                                | 1.29                                        | 3.82                                       |
| Document Paper | 340      | 0.15      | 9.02                                | -0.47                                       | 346.05                                     |

The items with the highest mean values shared the sense of immersion and presence provided by the environment. These

questions have in common the expressiveness of three-dimensional objects, avatars, and the surrounding environment constructed in the virtual world, as well as the physical and emotional communion that can be experienced through interaction with these objects, creating a high sense of immersion and presence. This phenomenon is caused by the fact that students are separated from the real world by a virtual space that is stacked up with them as a reference, and they can concentrate only on the wholly given environment, feeling as if there is no difference from the phenomena that occur in the real world. Additionally, all users who participated in the metaverse environment experienced less inequality in the same environment due to social factors such as gender, age, and job title. Therefore, suitability and fairness can be evaluated more strictly through virtual avatars created using anonymous information within the virtual environment.

Table 3: Mean of the Immersion evaluation questionnaire.

| Item  | Mean | 95% CI mean<br>lower bound | 95% CI mean<br>upper bound | Standard<br>deviation |
|-------|------|----------------------------|----------------------------|-----------------------|
| Total | 4.51 | 4.34                       | 4.68                       | 0.53                  |

### 3 CONCLUSION

We proposed a metaverse environment-based engineering education method that fulfills the potential as an ESG platform. We explain the overall taxonomy of ESG and the corresponding characteristics of the metaverse-based educational environment based on each detail and quantify the carbon footprint and social interaction elements to validate the proposed method. 340 students participated in the conference conducted for the carbon footprint measurement experiment, and if the conference was conducted offline, the carbon footprint that could occur could be 8,011 kgCO<sub>2</sub>eq by transportation and 346.05 kgCO<sub>2</sub>eq due to the use of disposable products. We have potentially reduced carbon emissions by 8,481 kgCO<sub>2</sub>eq by moving these seminars to a metaverse-based online environment. Immersion was measured using the Likert 5-scale through the IEQ, and 40 students participated in the seminar held for the experiment. As a result of the experiment, the average value of all items was approximately 4.51, indicating that the metaverse environment was separated from the real world due to the high sense of immersion and presence that the environment provided and experienced less inequality due to social factors. The experimental results confirmed that the metaverse-based educational environment presents students with new and interesting opportunities through an immersive environment-based educational method and has considerable strength as an ESG platform.

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